

BioFlow Guardian

Hack am Rhein Hackathon | Hossam Elshahaby | 3 October 2026

Problem

Basel is a major life-sciences manufacturing region, but external disruption signals often stay outside the factory decision loop. Rhine conditions, road congestion, and weather can affect refrigerated materials before production teams see a clear operational risk.

Solution

BioFlow Guardian converts public disruption data into four manufacturing actions: Buffer, Expedite, Reroute, and Quarantine. It gives operators a risk score, an action recommendation, and a plain-language explanation.

Decision Logic

Signal	Risk Question	Decision Impact
Rhine water level and discharge	Could logistics reliability deteriorate?	Expedite or Reroute
Basel traffic counts	Will the material spend longer outside controlled storage?	Buffer, Expedite, or Reroute
Weather observations	Does heat or storm risk increase cold-chain exposure?	Expedite, Reroute, or Quarantine
Cold-chain guidance	Does uncertainty require quality review?	Quarantine

Prototype Features

- Operator dashboard with current action, risk score, and production deadline.
- Scenario replay for normal day, heat plus traffic delay, Rhine disruption, and combined crisis.
- Explainable action cards that show why the system recommends Buffer, Expedite, Reroute, or Quarantine.
- Data-source panel for Rhine, traffic, weather, and cold-chain guidance.

Data Sources

Dataset	URL
Basel Rhine Water Level & Discharge	https://data.bs.ch/explore/assets/100089/
Basel Motor Traffic Counts	https://data.bs.ch/explore/assets/100006/
MeteoSwiss Open Data	https://opendatadocs.meteoswiss.ch/
MeteoSwiss Weather Stations	https://opendatadocs.meteoswiss.ch/a-data-groundbased/a1-automatic-weather-stations
Basel-Binningen Climate Data	https://data.bs.ch/explore/assets/100254/

Dataset	URL
Port of Switzerland Rhine Levels	https://port-of-switzerland.ch/hafenservice/pegel/
WHO Cold-Chain Guidance	https://www.who.int/publications/m/item/trs961-annex9-modelguidanceforstoragetransport
Rhine Suspended Solids Substances	https://data.bs.ch/explore/assets/100068/

Demo Scenario

A refrigerated oncology reagent is needed in six hours. The dashboard detects elevated traffic, high ambient temperature, and deteriorating Rhine conditions. BioFlow Guardian raises the risk score and recommends Reroute. If handling uncertainty increases, it recommends Quarantine for quality review.

Impact

The prototype helps operators act before external disruption reaches the production floor. It can reduce avoidable delays, improve cold-chain awareness, support quality review decisions, and make Basel's public data useful for manufacturing resilience.

Build Scope

MVP	Dashboard, data-source cards, risk scoring, scenario replay, and decision explanations.
Next	Live API ingestion, OpenStreetMap route optimization, material-specific stability profiles, and alerting.
Judging focus	Real Basel data, working decision logic, credible manufacturing scenario, and clear operator value.